2.1

a.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
−1	1	−2	4	−1	2
0	0	−1	1	−2	2
$\Sigma x_i = 5$	$\Sigma y_i = 10$	$\Sigma(x-\bar{x}) = 0$	$\Sigma(x-\bar{x})^2 = 10$	$\Sigma(y-\bar{y}) = 0$	$\Sigma(x_i-\bar{x})(y-\bar{y}) = 8$

$$\bar{x} = \frac{3+2+1-1+0}{5} = 1 \qquad \bar{y} = \frac{4+2+3+1+0}{5} = 2$$

b.

$$b_2 = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\Sigma(x_i-\bar{x})^2} = \frac{8}{10} = 0,8 \Rightarrow b_2 = 0,8$$ is the slope of fitted regression.

$$b_1 = \bar{y} - b_2\bar{x} = 2 - 0,8.1 = 1,2 \Rightarrow b_1 = 1,2$$ is the intercept of fitted regression.

$$\Rightarrow \hat{y} = 1,2 + 0,8x$$

c.

$$\Sigma_{i=1}^{5} x_i^2 = 3^2+2^2+1^2+(-1)^2+0^2 = 15$$

$$\Sigma_{i=1}^{5} x_i y_i = 3.4 + 2.2 + 1.3 + (-1).1 + 0.0 = 18$$

$$\left.\begin{array}{l}\Sigma(x_i-\bar{x})^2 = 4+1+0+4+1 = 10 \\ \Sigma x_i^2 - N\bar{x}^2 = 15 - 5.1^2 = 10\end{array}\right\} \Rightarrow \Sigma(x_i-\bar{x})^2 = \Sigma x_i^2 - N\bar{x}^2$$

$$\left.\begin{array}{l}\Sigma(x_i-\bar{x})(y_i-\bar{y}) = 4+0+0+2+2 = 8 \\ \Sigma x_i y_i - N\bar{x}\bar{y} = 18 - 5.1.2 = 8\end{array}\right\} \Rightarrow \Sigma(x_i-\bar{x})(y_i-\bar{y}) = \Sigma x_i y_i - N\bar{x}\bar{y}$$

d.

$$\hat{y} = 1,2 + 0,8x$$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3,6	0,4	0,16	1,2
2	2	2,8	−0,8	0,64	−1,6
1	3	2	1	1	1
−1	1	0,4	0,6	0,36	−0,6
0	0	1,2	−1,2	1,44	0
$\Sigma x_i = 5$	$\Sigma y_i = 10$	$\Sigma\hat{y}_i = 10$	$\Sigma\hat{e}_i = 0$	$\Sigma\hat{e}_i^2 = 3,6$	$\Sigma x_i\hat{e}_i = 0$

$$\bar{x} = \frac{3+2+1-1+0}{5} = 1$$

$$\bar{y} = \frac{4+2+3+1+0}{5} = 2$$

$$s_y^2 = \frac{\Sigma_{i=1}^{N}(y_i-\bar{y})^2}{N-1} = \frac{2^2+0^2+1^2+(-1)^2+(-2)^2}{4} = 2,5$$

$$s_x^2 = \frac{\Sigma_{i=1}^{N}(x_i-\bar{x})^2}{N-1} = \frac{2^2+1^2+0^2+(-2)^2+(-1)^2}{4} = 2,5$$

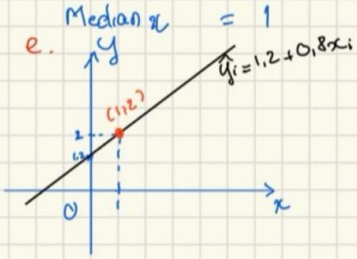
$$s_{xy} = \frac{\Sigma_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})}{N-1} = \frac{2.2+0.1+1.0+(-1).(-2)+(-2).(-1)}{4} = 2$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2,5}.\sqrt{2,5}} = 0,8$$

$$CV_x = 100\frac{s_x}{\bar{x}} = 100.\frac{\sqrt{2,5}}{1} = 158,11$$

Median x = 1

e.



f. The fitted line passes through $(1;2)$ point.

g. $b_1 + b_2\bar{x} = 0,8 + 1,2.1 = 2 = \bar{y}$

h. $\hat{y} = \frac{\Sigma\hat{y}_i}{N} = \frac{10}{5} = 2 = \bar{y}$

i. $\hat{\sigma}^2 = \frac{\Sigma(y_i-\hat{y}_i)^2}{N-2} = \frac{\Sigma\hat{e}_i^2}{N-2} = \frac{3,6}{3} = 1,2$

j. $\widehat{var}(b_2|x) = \frac{\hat{\sigma}^2}{\Sigma(x_i-\bar{x})^2} = \frac{1,2}{10} = 0,12 \Rightarrow se(b_2) = \sqrt{0,12} = 0,35$

2.22

a. *TOTALSCOREi* = 916.442 + 12.175x*SMALLi* + *ei*

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.675 249.396   <2e-16 ***
small         12.175      5.369   2.268   0.0236 *  
---
```

B1 = 916.442 shows that students in regular-sized classes have an average score of about 916.4

B2 = 12.175 shows that students in small classes score about 12.2 points higher on average than those in regular classes.

-> Reducing class size improves student learning outcomes.

b. *READSCOREi* = 432.665 + 6.924x*SMALLi* + *ei*

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.488 290.760  < 2e-16 ***
small          6.924      2.174   3.185  0.00151 ** 
---
```

MATHSCOREi = 483.777 + 5.251x*SMALLi* + *ei*

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.449 197.545   <2e-16 ***
small          5.251      3.578   1.467    0.143    
---
```

Students in small classes score about 6.9 points higher in reading, and this effect is statistically significant.

The effect on math scores is about 5.3 points but is not statistically significant.

-> Smaller class sizes improve reading performance more clearly than math performance.

c. *TOTALSCOREi* = 916.442 + 4.306x*AIDEi* + *ei*

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.559 257.529   <2e-16 ***
aide           4.306      4.994   0.862    0.389    
---
```

B1 = 916.442 shows that students in regular-sized classes without an aide have an average total score of 916.442.

B2 = 4.306 shows that students in regular-sized classes with a teacher aide score about 4.3 points higher on average but is not statistically significant.

-> No evidence that adding a teacher aide to a regular-sized classroom improves student learning.

d. *READSCOREi* = 432.665 + 2.871x*AIDEi* + *ei*

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.460 296.341   <2e-16 ***
aide           2.871      2.049   1.401    0.161    
---
```

. *MATHSCOREi* = 483.777 + 1.435x*AIDEi* + *ei*

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.355 205.464   <2e-16 ***
aide           1.435      3.304   0.434    0.664    
---
```

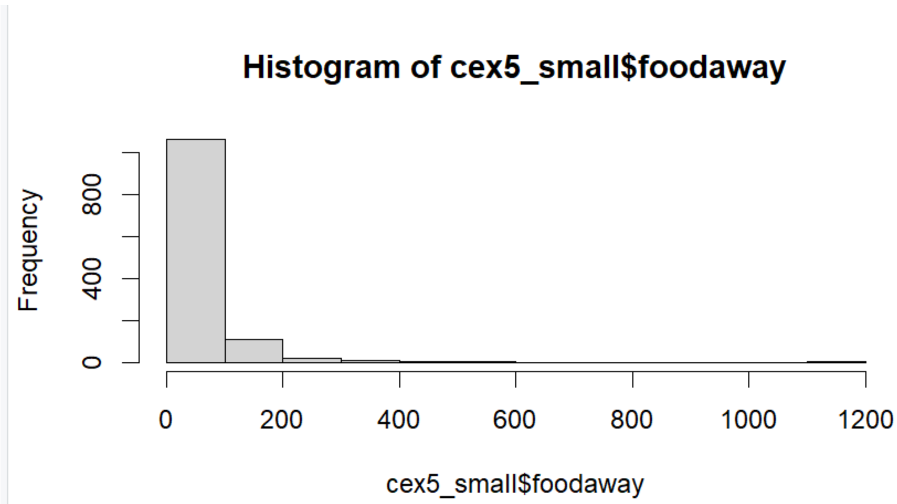
The coefficients on AIDE are positive, indicating that students in classes with a teacher aide score slightly higher on average. However, both effects are statistically insignificant.

-> Similar to part (c), there is no strong evidence that adding a teacher aide improves either reading or math performance.

2.25

a. Median = 32.55; mean = 49.27; 25^{th} percentiles = 12.04; 75^{th} percentiles = 67.5

```
> summary(cex5_small$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00   12.04   32.55   49.27   67.50 1179.00 
```



b.

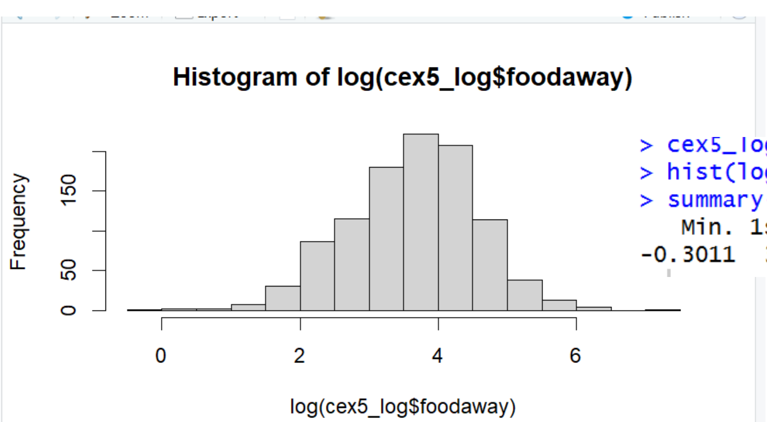
```
> advanced <- cex5_small[cex5_small$advanced == 1, ]
> summary(advanced$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00   21.67   48.15   73.15   90.00 1179.00 
> college <- cex5_small[cex5_small$college == 1, ]
> summary(college$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00   14.44   36.11   48.60   68.67  416.11 
> none <- cex5_small[cex5_small$advanced == 0 & cex5_small$college == 0, ]
> summary(none$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00    9.63   26.02   39.01   52.65  437.78 
```

Advanced: mean = 73.15; median = 48.15

College: mean = 48.60; median = 36.11

None of the above: mean = 39.01; median = 26.02

c.




```
> cex5_log <- cex5_small[cex5_small$foodaway > 0, ]
> hist(log(cex5_log$foodaway))
> summary(log(cex5_log$foodaway))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
-0.3011  3.0759  3.6865  3.6508  4.2797  7.0724 
```

ln(*FOODAWAY*) has fewer numbers of observations than *FOODAWAY* because we exclude any *FOODAWAY* =0 to take logarithm.

d.

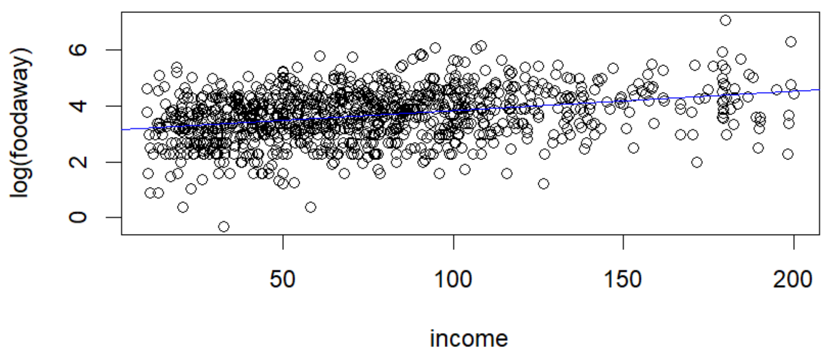
```
Call:
lm(formula = log(foodaway) ~ income, data = cex5_log)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.6547 -0.5777  0.0530  0.5937  2.7000 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
```

The slope coefficient indicates that a one-unit increase in INCOME leads to a 0.69% increase in food-away-from-home expenditure per person. The coefficient is statistically significant, suggesting that higher household income leads to greater spending on food away from home.

e. The scatter plot shows a positive relationship between income and log(FOODAWAY), and the fitted line slopes upward.



f. The residuals appear to be randomly distributed around zero with no obvious pattern, although the spread may increase slightly at higher income levels.